

FAIRFIELD, IA, USA

WMO: 726498

Lat: 41.053N

Lon: 91.979W

Elev: 799

StdP: 14.28

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) 1.1	(d) -11.5	(e) 3.0	(f) -2.5	(g) -7.6	(h) 3.8	(i) 2.5	(j) 29.3	(k) 17.4	(l) 26.9	(m) 21.4	(n) 8.7	(o) 300	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.8	(c) 93.2	(d) 77.5	(e) 90.7	(f) 76.6	(g) 88.3	(h) 75.2	(i) 80.7	(j) 90.3	(k) 78.7	(l) 87.7	(m) 77.0	(n) 85.7	(o) 9.3	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	151.7	87.8	75.5	137.6	84.7	73.4	127.9	83.0	45.0	90.5	42.9	87.2	40.8	85.5	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.2	(b) 22.1	(c) 19.4	(e) -10.6	(f) 97.6	(g) 6.5	(h) 4.3	(i) -15.3	(j) 100.7	(k) -19.1	(l) 103.3	(m) -22.8	(n) 105.7	(o) -27.5	(p) 108.8		
(4) 25.2	(5) 22.1	(6) 19.4	(7) -12.0	(8) 83.2	(9) 6.8	(10) 1.9	(11) -16.9	(12) 84.6	(13) -20.9	(14) 85.7	(15) -24.7	(16) 86.7	(17) -29.7	(18) 88.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	24.4	27.2	39.9	52.2	63.1	72.4	75.9	73.5	66.3	53.7	41.2	29.0	(6)
(7)		DBStd	20.56	12.27	12.29	11.76	10.13	8.53	6.25	6.08	5.85	8.26	9.64	10.64	11.81	(7)
(8)		HDD50	2915	794	642	353	91	6	0	0	2	71	300	656		(8)
(9)		HDD65	5936	1258	1058	779	397	141	12	5	81	371	715	1117		(9)
(10)		CDD50	3532	0	3	41	158	412	672	803	727	491	185	36	4	(10)
(11)		CDD65	1082	0	0	3	14	82	234	340	268	120	20	1	0	(11)
(12)		CDH74	10553	0	0	20	190	770	2193	3552	2456	1175	192	5	0	(12)
(13)		CDH80	3968	0	0	1	41	225	807	1520	925	408	41	0	0	(13)
(14)	Wind	WSAvg	9.1	10.3	10.3	10.9	11.3	9.9	8.1	6.5	6.0	7.1	8.8	10.0	9.8	(14)
(15)	Precipitation	PrecAvg	37.4	1.5	1.2	2.5	3.6	4.7	4.1	4.8	4.1	4.0	3.1	2.2	1.7	(15)
(16)		PrecMax	57.2	3.5	4.6	5.6	7.6	12.4	8.3	14.6	10.4	10.9	9.5	6.4	5.7	(16)
(17)		PrecMin	19.6	0.1	0.2	0.3	0.7	1.4	0.9	0.2	0.9	0.1	0.0	0.0	0.1	(17)
(18)		PrecStd	7.8	0.9	0.9	1.5	1.7	2.5	2.1	3.4	2.2	2.5	2.1	1.5	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	57.2	63.9	77.2	85.0	90.8	93.3	99.2	96.9	92.9	84.3	74.6	63.2		(19)
(20)		MCWB	47.4	52.4	62.4	67.1	73.3	77.6	78.2	74.4	75.3	69.9	60.7	57.2		(20)
(21)		DB	50.3	55.1	69.9	79.4	85.9	90.7	94.7	91.4	89.6	79.2	68.0	54.9		(21)
(22)		MCWB	44.5	47.5	59.3	62.5	71.1	76.3	79.7	77.7	73.3	64.7	57.0	49.9		(22)
(23)		DB	45.0	48.3	63.5	74.7	81.7	88.3	90.9	88.5	84.4	74.7	63.0	49.6		(23)
(24)		MCWB	40.0	43.2	55.2	60.4	67.9	74.6	77.7	76.7	71.8	62.8	54.5	45.0		(24)
(25)		DB	39.2	43.3	57.3	70.1	78.6	84.5	88.2	85.5	81.2	70.0	57.3	44.9		(25)
(26)		MCWB	35.5	38.8	49.3	57.4	65.8	72.7	76.2	74.7	68.6	60.6	49.8	40.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	50.1	54.8	65.3	69.0	76.5	82.3	83.5	83.0	78.4	72.8	63.1	59.2		(27)
(28)		MCDWB	53.5	62.7	74.6	80.3	86.5	90.6	94.1	91.6	88.2	82.9	72.2	63.0		(28)
(29)		WB	44.3	48.7	60.2	65.5	73.0	78.4	81.0	80.2	75.7	68.6	59.0	51.5		(29)
(30)		MCDWB	49.5	55.3	66.8	77.1	82.8	87.3	91.6	88.9	85.4	75.1	65.5	54.3		(30)
(31)		WB	39.9	43.5	55.9	62.2	70.5	76.8	79.2	77.9	73.4	65.0	55.2	45.3		(31)
(32)		MCDWB	45.0	48.6	62.4	71.9	79.7	85.7	88.7	86.3	82.3	71.7	62.1	48.4		(32)
(33)		WB	36.2	39.0	50.2	59.2	67.8	74.5	77.4	75.6	71.2	61.4	50.5	40.4		(33)
(34)		MCDWB	39.4	43.2	57.4	68.5	75.9	82.7	86.1	83.0	78.7	69.0	57.2	44.3		(34)
(35)	Mean Daily Temperature Range	MDBR	16.0	16.7	19.1	21.9	19.7	19.1	19.8	20.0	22.5	20.9	18.5	15.2		(35)
(36)		5% DB	21.5	23.9	26.1	28.1	23.4	21.8	21.9	22.8	24.1	25.7	24.6	20.3		(36)
(37)		MCWBR	16.4	17.3	16.8	16.1	12.6	11.4	11.0	12.4	12.0	14.1	16.2	16.0		(37)
(38)		5% WB	21.4	22.5	23.5	24.9	20.5	19.5	19.6	19.9	20.5	21.1	21.7	19.6		(38)
(39)		MCWBR	16.5	17.0	16.4	16.0	12.5	12.3	11.6	12.6	11.5	14.2	16.1	16.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.295	0.307	0.337	0.385	0.418	0.420	0.440	0.423	0.389	0.343	0.300	0.290		(40)
(41)		taud	2.421	2.384	2.392	2.266	2.244	2.284	2.215	2.279	2.324	2.456	2.508	2.451		(41)
(42)		Ebn at Noon	274	287	290	281	273	272	265	266	268	271	271	266		(42)
(43)		Edh at Noon	25	30	34	41	43	42	44	40	36	28	23	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	592	901	1189	1507	1766	1965	1967	1727	1477	984	664	498		(44)
(45)		RadStd	65	72	96	115	126	194	137	99	109	110	56	55		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page